Excitation-Aware On-Track System Identification at Full Scale: An Identifiability Analysis and a Simulation Architecture for Pacejka Estimation

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***Abstract*—On-track system identification promises to learn tire models from ordinary racing laps instead of dedicated steady-state manoeuvres. We study the identifiability of one such residual-learning pipeline at full vehicle scale, on a simulated plant that publishes its own per-wheel slip angles, loads and lateral forces, so the estimator can be scored against the forces the plant actually generated. Transplanted unchanged from a small-scale platform, the pipeline converges on a tire with less than half the vehicle's peak grip and most Pacejka coefficients resting on their box constraints. This is not a solver, bounds or tuning failure but a loss of excitation created by the pipeline's own virtual-data step: the friction the synthetic rollout can demand, $\mu_{\text{reach}} = v^2\delta_{\text{sweep}}/(gL)$, is a property of the rollout configuration, not of the tire. We contribute this identifiability bound and a rollout that sets its speed from the measured lap; a bounded steering sweep and a frozen regularisation target, together removing a 1.01→1.87 peak-factor drift over six co-identification iterations; measurement corrections—achieved rather than commanded road-wheel angle, measured mass and wheelbase—and a friction warm start read from the *curvature* of the axle force curve and applied only as a floor; admissibility gates on derived physical quantities at the model-to-controller boundary; and an exact restructuring of the S4D temporal residual—closed-form kernel, hoisted physics-loss terms, rollout on a CPU—that cuts one identification from 181 s to 21 s without moving a coefficient. After correction, no coefficient rails, peak friction and axle cornering stiffnesses land within 20 % of limits added.**

***Index Terms*—System identification, tire modeling, Pacejka Magic Formula, identifiability, persistent excitation, vehicle dynamics, autonomous racing, simulation architecture.**

I. Introduction

Model-based controllers for autonomous racing derive their performance from the fidelity of the vehicle model they carry [1], [2], [3]. In the lateral channel that fidelity is dominated by the tire, which is simultaneously the physical bound on manoeuvrability and the least accessible component to measure: its behaviour depends on surface, temperature, wear and load in ways that change between sessions [4], [5].

Classical characterisation resolves this by removing the vehicle from the track—steady-state circular or ramp-steer manoeuvres in a large open space yield clean data at a logistical cost a race weekend rarely affords [6]. On-track identification removes the logistics but reintroduces transients, noise and, critically, the fact that a racing line is a poor excitation signal for a nonlinear model; nonlinear least squares applied directly to lap data is known to be brittle in exactly this regime [6].

Dikici *et al.* [6] propose the hybrid remedy that is the starting point of this paper. A small residual network is trained on 30 s of ordinary lap data to predict the one-step error of a nominal single-track model; the corrected model is then rolled out through a *synthetic*, quasi-steady steering sweep; and Pacejka coefficients are fitted to that synthetic sweep. Because the sweep is generated rather than driven it can be made steady and noise-free, and because the residual network only has to interpolate within its training distribution its generalisation weakness is contained. The procedure iterates, each fit becoming the next nominal model. On a 1:10 platform it reports a 3.3× lower one-step RMSE than nonlinear least squares under noise, from 30 s of driving [6].

A. Problem statement

This paper asks what the pipeline identifies when it is moved from a 3.7 kg, 0.32 m-wheelbase platform to a full-scale Formula Student vehicle: 240.0 kg, 1.53 m wheelbase, 16 ° of road-wheel lock, driven at 11–27 m/s. The question is one of identifiability rather than of parameter transfer. Two scale-dependent mechanisms, both latent at 1:10, break the estimate:

1) **Excitation collapses inside the virtual-data step.** The Pacejka fit never sees the recorded lap; it sees the synthetic rollout, whose reachable friction $\mu_{\text{reach}} \approx v^2\delta_{\max}/(gL)$ is a property of the rollout configuration, not of the tire. A rollout speed inherited from the scaled platform caps it at 0.43 here. Below the tire's peak the Magic Formula degenerates to $F_y \approx F_z(BCD)\alpha$: only the product BCD is identifiable and the bounded optimiser resolves the remaining freedom on a box edge [7], [4].
2) **Lateral demand scales as v^2, so a bad grip estimate is silent until it is catastrophic.** An identified peak friction coefficient of 0.40 produces 0.03 m of cross-track error at 11 m/s and 41.89 m at 27 m/s. The scaled platform never reaches the speed at which the error becomes observable.

A third difference is an opportunity. Because the study is conducted in the CARLA simulator [8] through a purpose-built

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ROS 2 interface, the simulator's own per-wheel telemetry—slip angle, normal load, lateral force, per wheel—is available as ground truth, so an on-track identifier can be scored against the forces the plant actually generated.

B. Contributions

The contributions are an identifiability result, the pipeline changes it implies, and the simulation architecture that makes both verifiable.

1) **An identifiability analysis of the virtual-data step** (Section V-B), giving the closed-form reachable-friction bound above and an *excitation-aware rollout* that selects the rollout speed from the measured lap and warns whenever the configuration cannot reach the prior's peak. This corrects the baseline method [6] at any scale; it is latent at 1:10 because a 2–4 m/s rollout is representative there and is not here.
2) **Two mechanisms that stabilise the iterative co-identification** (Section V-C): the synthetic sweep is bounded at $1.5\times$ the 99th percentile of the observed steering, so the fit cannot read the residual network's extrapolation as grip; and the Tikhonov target is frozen at the configured prior while the optimiser's start point tracks the iterations. Together they remove a $1.01\to1.87$ drift of the peak factor over six iterations.
3) **Measurement-side corrections for full-scale deployment** (Section V-E): slip angles built from the *achieved* road-wheel angle rather than the command; mass and wheelbase from measured static wheel loads rather than from a configuration file; and a friction warm-start computed as a high quantile over the whole buffer and applied as a *floor* rather than as a median over a low-slip mask.
4) **A simulation architecture for on-track identification research** (Sections IV-B–IV-D). CARLA_ASU_BRIDGE exposes the plant's own per-wheel forces, slip angles and loads as ROS 2 topics alongside a conventional autonomy interface, and closes the identification loop end to end—acquisition, preprocessing, fitting, ground-truth force validation—on a single simulation clock.
5) **Admissibility gates on derived physical quantities** (Section VII) at the model-to-controller boundary—axle cornering stiffness, understeer sign and critical speed, and grip ceiling against the trajectory's lateral demand—rather than per-coefficient box checks, which cannot detect a degenerate fit or a pairwise-oversteering axle set.
6) **A negative result reported rather than suppressed** (Section VI-J). The corrected model reports a grip ceiling of 1.02–1.16 g against a measured steady-state ceiling of $\approx 1.0\,g$, while the optimised raceline used here demands $1.22\,g$; the gate refuses the identification for closed-loop use, and we quantify the raceline regeneration this implies instead of loosening the gate.

The paper follows that order: Section II places the work against the state of the art; Section III states the identification problem and its structural identifiability properties; Section IV describes the simulation architecture and data pipeline; Section V the identification methodology; Section VI the results; Section VII how the identified model is handed to the controller; and Sections VIII–IX the limitations and outlook. All results are obtained in simulation, and Section VIII states which conclusions are expected to transfer to hardware.

II. Related Work and Position

A. Tire models and Magic Formula identification

The Magic Formula of Bakker, Nyborg and Pacejka [9] and its consolidation [10], [4] remains the standard empirical description of steady-state tire force generation and is the model identified here; its lateral form and the two structural properties this paper relies on are stated in Section III. Classical characterisation obtains the coefficients from steady-state manoeuvres in a controlled space [6], [5]: clean data at high logistical cost. Direct fitting to measured contact-patch forces is the most informative variant and requires a force-measuring hub or a simulator's own wheel telemetry; in this work it is used only as a *reference* against which the on-track identifier is scored, never inside the identification loop.

B. On-track and data-driven identification

On-track methods trade excitation quality for operational simplicity. Nonlinear least squares on lap data is the classical baseline and is brittle under noise [6]. Gaussian-process approaches learn the mismatch of a nominal model and have been used both for identification and inside the controller [11], [12], [2]; they are accurate but carry a continuing computational cost and, as [6] observes, are usually evaluated starting from an already accurate physical model. Neural residuals replace the Gaussian process with a cheaper regressor at the cost of generalisation guarantees.

Deep Dynamics [13] is the closest work on the physics-constraint axis: it estimates Pacejka coefficients inside a neural network and adds a *Physics Guard* layer clamping the internal estimates into nominal ranges. Our admissibility gates share the motivation but differ in placement and in what is constrained. We report as a substantive finding that per-coefficient box constraints are *insufficient*: the degenerate fit that motivated this work satisfied every box constraint while five of eight coefficients sat exactly on a bound, and a later fit satisfied every box constraint while being pairwise oversteering and open-loop unstable above 17.4 m/s. We therefore gate on derived quantities at the model-to-controller interface and treat a coefficient resting on a bound as a diagnostic of non-identifiability rather than as a successful clamp.

The hybrid residual-plus-virtual-data structure of [6] is the pipeline we extend. Its key idea—query the learned model only where it interpolates, and extract physically parameterised coefficients from it—is retained unchanged. What we correct is the mechanism that decides *where* the learned model is queried.

C. Overlap with concurrent work

Wu *et al.* [14] extend the same baseline concurrently and independently, and the overlap must be stated plainly. They introduce (i) a vision-based friction prior from a lightweight CNN over road texture, used to warm-start the optimisation; (ii) an S4 structured state-space model in place of the MLP; and (iii) Nelder–Mead for the iterative Pacejka extraction, with virtual simulation in CarSim. The pipeline described here independently contains an S4D temporal residual [15], [16], a friction warm-start, and Nelder–Mead and differential-evolution solver options alongside trust-region reflective. We therefore do *not* claim the temporal-residual architecture, the friction warm-start concept, or the derivative-free solver as contributions.

Three differences are technical rather than chronological. First, our friction prior is proprioceptive—derived from measured accelerations, in the spirit of model-free friction estimation from inertial data [17]—and therefore needs no camera or textured-surface assumption. Second, a prior computed from *utilised* friction is a lower bound on *available* friction and cannot substitute for excitation: on a gentle lap the median utilisation on our vehicle measures 0.044. We therefore identify the peak from the curvature of the axle force curve, by a gated two-parameter brush-model fit [18] of the kind surveyed by Acosta *et al.* [19], and—whatever the fit's own confidence—apply the result only as a floor, for a reason that belongs to the interface with the controller rather than to the estimator (Section VI-F1). The actual fix for the degeneracy is located in the rollout excitation, not in the initialisation. Third, we report the cost of the temporal residual and reduce it by 8.7× without changing its output (Section V-A3), which is what makes an S4-family residual a deployable option on a vehicle CPU rather than an offline one; the techniques are standard for diagonal state-space models [15], [16], [20] and we claim only the measurement and the exactness. To our knowledge the reachable-friction bound of Section V-B and its consequence for the virtual-data step have not previously been stated.

D. Simulation architectures for identification and validation

CARLA [8] is the standard open simulator for autonomous-driving research and supplies the sensor and scenario infrastructure used here. Its native dynamics come from the underlying PhysX wheeled-vehicle model—a rigid-body model with per-wheel tire friction rather than an empirical Magic Formula fit to measured data—which is why this study treats the simulator as a well-instrumented *plant* to be identified and not as a source of ground-truth Pacejka coefficients. Recent work addresses that fidelity gap directly: R-CARLA [21] makes the dynamics model interchangeable while retaining CARLA's sensor simulation, and Sagmeister *et al.* [22] quantify how far a vehicle model can be simplified before conclusions about a trajectory-following controller stop transferring—finding, relevantly here, that sensitivity is largest near the acceleration limits. CARLA has also hosted complete Formula Student Driverless stacks with ROS 2 interfaces [23]. The architecture of Section IV sits in this line and is distinguished by exposing the simulator's own per-wheel slip, load and force telemetry on a simulation-time-stamped interface, so that an identification method can be scored against the forces the plant actually produced rather than against a secondary model. Scaled platforms—F1TENTH [24] and the ForzaETH Race Stack [3]—provide the software context of the baseline; full-scale Formula Student Driverless systems [5] the vehicle class targeted here.

E. Downstream consumer and gap addressed

The identified model is consumed by a nonlinear MPC path-tracking controller built on `acados` [25] with partial-condensing HPIPM [26], following standard practice in racing MPC [1], [2], tracking a minimum-curvature raceline generated in the manner of Heilmeier *et al.* [27]. We claim no contribution to the controller itself; the controller appears in this paper only as the consumer that defines what an admissible identification is (Section VII).

Summarising the gap: the baseline [6] establishes that a residual network plus virtual steady-state data recovers a Pacejka model from a lap at 1:10 scale, and concurrent work [14] improves its residual architecture and its initialisation. Neither addresses the condition under which the virtual data itself carries information about the tire's peak, which is the binding constraint at full scale, and neither treats the admissibility of the identified model for its downstream controller as an explicit, physically derived contract. This paper addresses both, on a platform where the answer can be checked against the plant's own forces.

III. Identification Problem

A. Model structure

The identification model is the dynamic single-track model of the baseline, with mass m, yaw inertia I_z and centre-of-gravity distances l_f, l_r, $L = l_f + l_r$. Longitudinal and lateral dynamics are decoupled and load transfer is neglected:

$$\dot{v}_y = \tfrac{1}{m}\big(F_{y,r} + F_{y,f}\cos\delta - m v_x \omega\big), \tag{1}$$
$$\dot{\omega} = \tfrac{1}{I_z}\big(F_{y,f} l_f \cos\delta - F_{y,r} l_r\big), \tag{2}$$

with slip angles

$$\alpha_f = \delta - \arctan\!\Big(\tfrac{v_y + l_f\omega}{v_x}\Big), \quad \alpha_r = -\arctan\!\Big(\tfrac{v_y - l_r\omega}{v_x}\Big). \tag{3}$$

Axle forces follow the lateral Magic Formula

$$F_y = F_z\, D \sin\!\big(C \arctan\!\big(B\alpha - E(B\alpha - \arctan B\alpha)\big)\big), \tag{4}$$

with per-axle coefficients $\Phi_p = [B_f, C_f, D_f, E_f, B_r, C_r, D_r, E_r]$. The identification state and input are $\mathbf{x} = [v_y, \omega]$ and $\mathbf{u} = [v_x, \delta]$, and the nominal one-step map is $\hat{\mathbf{x}}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k, \Phi_p)$. The baseline discretises (1)–(2) by explicit Euler at T_s; we retain Euler as the default and expose second- and fourth-order Runge–Kutta as options, because the full-scale vehicle's lateral mode is stiffer relative to T_s than the scaled platform's.

B. Objective and derived quantities

The identification objective is to recover Φ_p from a single 30 s segment of ordinary lap data, without dedicated manoeuvres, such that the resulting model is both predictive and *admissible* for the controller that consumes it. Admissibility is expressed on derived quantities rather than on coefficients. The linearised axle cornering stiffness is the slope of (4) at zero slip,

$$C_i = F_{z,i} B_i C_i^{\mathrm{MF}} D_i, \qquad i \in \{f, r\}, \tag{5}$$

(writing C^{MF} for the shape factor to avoid a clash), and the grip ceiling implied by an identified set is

$$a_y^{\max} = \frac{D_f F_{z,f} + D_r F_{z,r}}{m}. \tag{6}$$

The linear model built from (5) is oversteering when $l_f C_f > l_r C_r$ and is then open-loop unstable above the critical speed $v_{\mathrm{crit}} = L\sqrt{C_f C_r / \big(m(l_f C_f - l_r C_r)\big)}$ [28].

C. Structural identifiability

Two properties of (4) are used throughout. First, its slope at zero slip is $F_z BCD$: the cornering stiffness depends on the coefficients only through that product, and E drops out. Second, and decisively, if the data does not approach the peak then (4) linearises to $F_y \approx F_z(BCD)\alpha$ and B, C, D become mutually unidentifiable. This is the standard persistent-excitation argument [7] and is not claimed as novel. What we claim, and demonstrate in Section V-B, is that the baseline pipeline's virtual-data step can silently violate it, and that a bounded optimiser then resolves the remaining freedom by sliding onto a box edge—so the violation is reported as a converged fit rather than as a failure.

D. Baseline pipeline

The baseline proceeds in four stages, repeated for N_{it} iterations [6]:

1) **Residual learning.** A small network f_{NN} is trained on the recorded lap to predict the one-step error $e_k = \mathbf{x}_{k+1} - \hat{\mathbf{x}}_{k+1}$ from $[v_x, v_y, \omega, \delta]_k$; the corrected model is $\mathbf{x}_{k+1} = \hat{\mathbf{x}}_{k+1} + e_k$.
2) **Virtual steady-state generation.** The corrected model is integrated open-loop at constant longitudinal speed while the steering angle is ramped from 0 to δ_{sweep}, producing a synthetic quasi-steady dataset.
3) **Pacejka fit.** Assuming $\dot{v}_y = \dot{\omega} = 0$ on the synthetic data, axle forces are recovered from the yaw balance,
$$F_{y,r} = \frac{m l_f v_x \omega}{L}, \qquad F_{y,f} = \frac{m l_r v_x \omega}{L \cos\delta}, \tag{7}$$
slip angles from (3), and Φ_p is fitted to $(\alpha_i, F_{y,i})$ by bounded least squares.
4) **Re-nomination.** The fitted Φ_p becomes the next iteration's nominal model and the residual network is reinitialised.

The structural point that drives the rest of this paper is that stage 3 *never sees the recorded lap*. It sees the output of stage 2. The recorded lap enters only through the residual network, so the excitation available to the fit is set by the rollout configuration and not by how the vehicle was driven.

IV. Simulation Architecture and Data Pipeline

The identification pipeline is exercised end to end inside a single simulation architecture: a Formula Student vehicle model in CARLA acting as the plant, a C++ ROS 2 communication layer that drives it and republishes its state, and an identification stack that consumes that state. The architecture is described here because two of its properties—per-wheel ground truth and simulation-time stamping—are what make the identifiability claims of Section V checkable rather than merely arguable.

A. Choice of simulator

CARLA [8] was selected against four requirements that the identification problem of Section III imposes on a simulator.

Per-wheel ground truth. The claim to be checked is that the recovered Magic Formula reproduces the lateral force the plant actually produced, so the simulator must expose per-wheel slip angle, normal load and tire force—not only chassis states. CARLA's PhysX wheeled-vehicle solver computes these quantities internally and its client API exports them, which is what makes the tire forces feedback channel of Section IV-C possible. Cone-oriented Formula Student simulators.

Deterministic, simulation-time-stamped execution. Persistent-excitation and residual-learning arguments are only checkable if every sample carries a consistent clock. CARLA's synchronous mode with a fixed physics step (0.033 s here) decouples data timing from wall-clock load; all identification data is stamped from the simulation clock (Appendix, Section A7).

Full-scale vehicle and racing-track support. The target class is a full-scale Formula Student car on a closed circuit. CARLA accepts custom vehicle assets here the Formula AI vehicle on the silverstone map and offers a client that the bridge links against directly, avoiding a Python round trip in the control loop. Surveys of open self-driving simulators place CARLA (with the now-discontinued LGSVL) at the state of the art for end-to-end stack testing on exactly these grounds [29].

A plant whose tire behaviour is not *a Magic Formula.* CARLA's native dynamics are a known fidelity gap: R-CARLA [21] replaces the dynamics model to shrink a measured sim-to-real gap, and Sagmeister *et al.* [22] quantify where model simplification changes control conclusions. For this study that gap is a feature rather than a defect. Because forces come from a friction-based rigid-body solver and not from an analytic tire model, the simulator has no built-in Pacejka coefficients for the identifier to trivially recover; the reference curve must itself be fitted to the plant's telemetry (Section VI), which is the same position the method occupies on a real vehicle. The limits of this choice are revisited in Section VIII.

B. Plant: Formula AI vehicle in CARLA

The plant is the driverless Formula Student–class vehicle model, instantiated in CARLA [8] (server 0.9.16, Unreal Engine 4.26) on the `silverstone` map: a four-wheeled, front-steered, all-wheel-drive rigid-body vehicle whose lateral

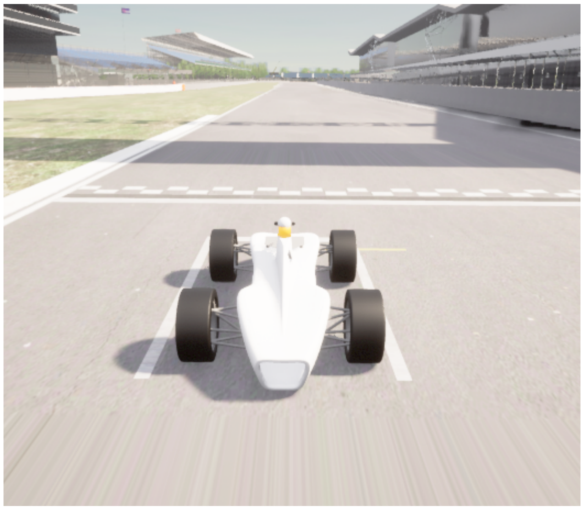

Fig. 1. The Formula AI vehicle model, on the start point of the silverstone map in CARLA. Its lateral forces are produced by the engine's PhysX wheeled-vehicle solver rather than by an analytic tire model, which is why the reference curve of Table **??** is fitted to the simulator's own per-wheel telemetry. Measured geometry and mass are given in Table I.

TABLE I
VEHICLE CONFIGURATION AND MEASURED PLANT PROPERTIES. CONFIGURED VALUES ARE SET IN THE BRIDGE'S `CARLA_INTERFACE_CONFIG.YAML`; MEASURED VALUES ARE OBTAINED FROM THE SIMULATOR'S PER-WHEEL TELEMETRY.

Quantity	Value	Unit
Mass m	240.0	kg
Static wheel load (total)	2644.5	N
Front/rear static split	51.2 / 48.8	%
Wheelbase L	1.5334	m
l_f / l_r	0.7381 / 0.7953	m
Yaw inertia I_z[a]	51.1	$\mathrm{kg\,m^2}$
Wheel radius	0.25	m
Max road-wheel angle	16.0	°
friction coefficient μ **default**	1.5	—
Drag coefficient	0.426	—

[a]Assumed; see Section A.

forces are generated by the PhysX wheeled-vehicle solver under a configured friction coefficient, not by an analytic Magic Formula. The simulator therefore has no "true" Pacejka coefficients to recover, and the reference tire curve of Section VI is itself a Magic Formula fit to the simulator's own per-wheel force, slip and load telemetry under a dedicated limit-excitation manoeuvre.

Three rows of Table I are load-bearing for the identification. First, the mass the physics engine simulates (240.0 kg, measured as total static normal load) is 12 % above the configured 240 kg; since (7) is linear in m, the configured value biases every recovered force by that amount. Second, the PhysX wheel material friction coefficient (1.5) is *not* the realised peak axle friction (1.00–1.05), so an identifier tuned to converge on the configuration file converges on the wrong target. Third, the achieved road-wheel angle is 0.76 of the commanded angle, so slip angles built from the command are systematically too large.

TABLE II
ROS 2 INTERFACE USED BY THE IDENTIFICATION AND VALIDATION PIPELINE. NAMESPACE IS `/SIM` UNLESS MARKED ABSOLUTE.

Topic	Type	Role
`/odom`	`nav_msgs/Odometry`	State v_x, v_y, ω; twist is body-frame
`feedback/steering_angle`	`std_msgs/Float32` (deg)	Achieved road-wheel angle δ
`feedback/imu`	`sensor_msgs/Imu`	a_x, a_y for the friction warm-start
`feedback/tire_forces`	`sim_manager_msgs/TireForces`	Per-wheel α, F_z, F_y: **ground truth**
`feedback/speed`	`std_msgs/Float32`	Forward speed echo
`feedback/motors`	`std_msgs/String` (JSON)	Per-wheel speed/torque/brake
`/drive`	`ackermann_msgs/AckermannDriveStamped`	Control command in
`control/tire_friction`	`std_msgs/Float32`	Runtime grip override
`/clock` *(abs.)*	`rosgraph_msgs/Clock`	Simulation time base

C. CARLA_ASU_BRIDGE*: interfaces and telemetry*

CARLA_ASU_BRIDGE [30] is a C++ ROS 2 lifecycle node linked directly against CARLA's client library (Fig. 2). It spawns and configures the ego vehicle, drives its inputs, and republishes simulator state as standard ROS 2 topics; all identification traffic crosses this interface.

1) Command path: The bridge runs in `ackermann_drive` mode, subscribing to a single `ackermann_msgs/AckermannDriveStamped` on `/drive` and forwarding speed and steering angle to CARLA's native Ackermann controller, whose server-side cascaded PID loops are configured once at startup. No client-side steering PID is present—the commanded road-wheel angle passes through in radians—so the achieved/commanded ratio of Table I is a property of the vehicle and its steering curve rather than of a tuning choice in the bridge.

2) Feedback path: Table II lists the interface consumed by the identifier and its validation. Three properties are specific to this architecture.

a) Per-wheel ground truth: `feedback/tire_forces` is a direct passthrough of CARLA's own `WheelTelemetryData`: per-wheel lateral slip angle, normal load, lateral force, longitudinal force and wheel torque, in wheel order [FL, FR, RL, RR]. No analytic tire model exists inside the bridge, so nothing on this topic can disagree with the physics the vehicle is driven by. Lateral force is sign-flipped and slip angle converted into the ROS right-handed body frame; the longitudinal channel is drivetrain torque at the axle ($-\tau/r_w$) rather than a contact-patch force and is unused here. This topic is what allows an on-track identifier to be scored against the forces the plant generated.

b) Simulation-time stamping: Every feedback message is stamped from the same simulation epoch as `/clock`. In synchronous mode CARLA ticks as fast as the server manages (observed near $1.9\times$ real time), so wall-clock stamping would drift the identification dataset against itself without bound.

c) Odometry and frame conventions: Odometry is computed from the actor transform and velocity at 50 Hz with `follow_server_rate` enabled and an optional additive

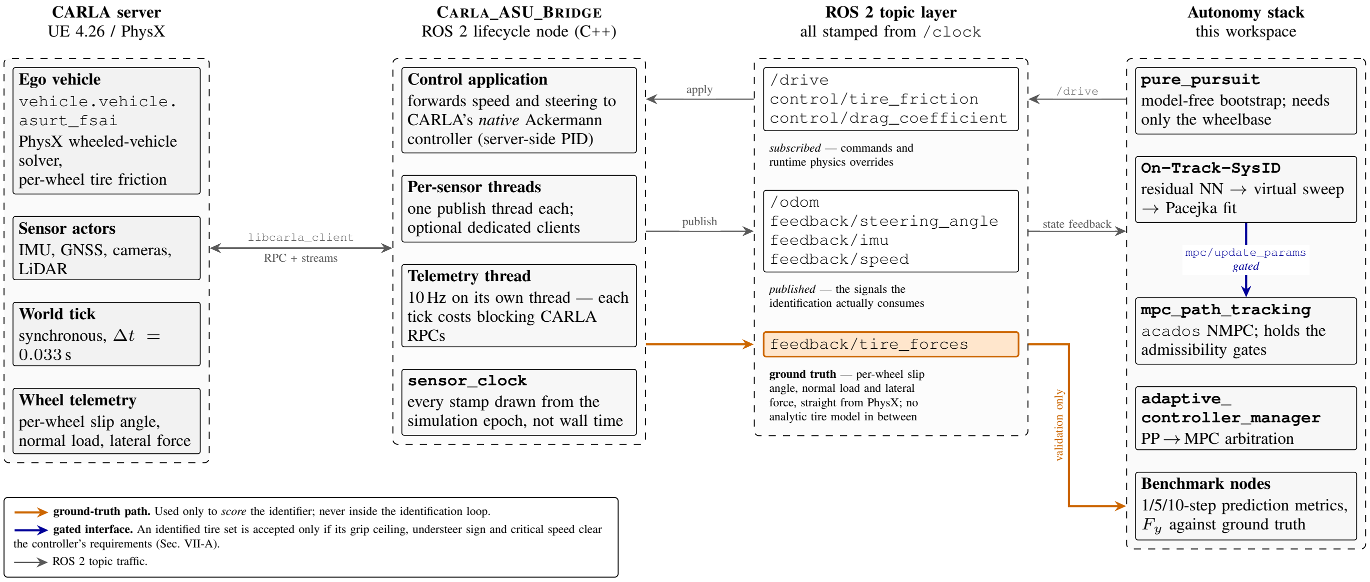

Fig. 2. Architecture of the CARLA_ASU_BRIDGE ROS 2 interface. The bridge links against CARLA's C++ client, drives the ego vehicle through CARLA's native Ackermann controller, and republishes state, sensors and per-wheel telemetry on ROS 2 topics stamped from the simulation clock. The `feedback/tire_forces` path (highlighted) carries the simulator's own per-wheel slip angle, normal load and lateral force, and is the ground truth against which the identifier is scored.

Gaussian noise model (disabled here). Publisher and subscribers all use `BEST_EFFORT`, because a reliability mismatch on this interface is silent: DDS requires requested $\leq$ offered, so a `RELIABLE` request against a `BEST_EFFORT` publisher yields one warning and then no data. Because (3) is meaningless in the wrong frame, the body-frame assumption on `/odom`'s twist was verified rather than assumed: correlation 0.93 with 0.09 m/s RMS error against the reconstructed body-frame lateral velocity, versus 16.7 m/s RMS under a world-frame interpretation.

3) Extensions made for identification: Four changes support this study and are listed for reproducibility: the existing but unsubscribed `feedback/steering_angle` is now consumed by the identifier (with configurable units, scale and staleness timeout, falling back to the `/drive` setpoint); runtime overrides `control/tire_friction` and `control/drag_coefficient` write straight to the live `VehiclePhysicsControl`, so surface friction can change within a run without respawning; the torque curve, zero-throttle damping and wheel `tire_friction` were tuned against a low-speed hold test (a wheel friction above $\approx$2.0 on a steered wheel scrubs hard enough at creep speed to stall the drivetrain intermittently), and these values define the plant being identified; and telemetry runs on a dedicated 10 Hz thread off the control loop, since each tick costs several blocking CARLA RPCs.

D. Data acquisition and preprocessing

The vehicle is driven around the reference raceline by a model-free pure-pursuit controller, which requires no tire knowledge—only the wheelbase—and is therefore a valid bootstrap, as in the baseline [6], [3]. Odometry, the achieved steering angle and the drive command are sampled into a ring buffer at 50 Hz ($T_s = 0.02$ s) for 30 s, gated on $v_x > 1$ m/s. Preprocessing follows the baseline: a zero-phase low-pass filter, then symmetry augmentation by mirroring $(v_y, \omega, \delta) \rightarrow (-v_y, -\omega, -\delta)$ at unchanged v_x, which doubles the data and removes the cornering bias of a circuit. Re-identification retriggers every 30 s by default.

E. How the architecture closes the loop

The four stages of the identification pipeline map onto the architecture as follows. *Acquisition* uses `/odom`, `feedback/steering_angle` and `/drive`, all stamped on the simulation clock, so that the sampling interval used by the one-step model is the interval the physics actually advanced. *Preprocessing* and *fitting* run inside the identification node against that buffer. *Validation* draws on `feedback/tire_forces` and `feedback/imu`, which never enter the identification loop, so the estimator is scored against a channel it did not observe. *Deployment* pushes the fitted coefficients over a ROS 2 service to the controller behind the gates of Section VII. Only the last stage leaves the architecture, which is what makes the negative result of Section VI-J a statement about the trajectory rather than about the interface.

V. Identification Methodology

This section describes the identification pipeline as it stands after correction. Section V-A covers the residual model, V-B–V-C the two changes to the virtual-data step that constitute the paper's central methodological contribution, V-D the fit itself, V-E the measurement chain, and V-H the validation protocol. Figure 3 summarises the loop.

A. Residual model

The default residual is the baseline multilayer perceptron: 4 inputs, one hidden layer of 8 units, 2 outputs, LeakyReLU,

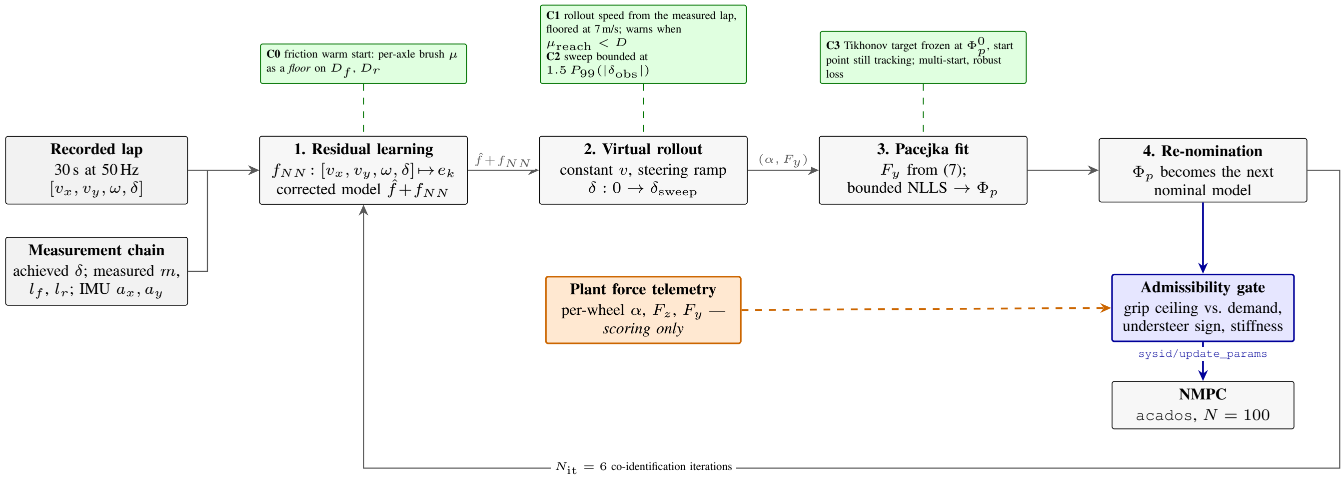

Fig. 3. The identification pipeline. Recorded lap data trains the residual network; the corrected model is rolled out through a synthetic steering sweep; Pacejka coefficients are fitted to that sweep; the fit becomes the next nominal model. Green marks this paper's changes to the inherited loop: the friction warm start applied as a floor on D (Section V-F), the excitation-aware rollout speed and the extrapolation-bounded sweep (Sections V-B–V-C), and the frozen regularisation target of (10). Blue is the gated model interface to the controller (Section VII); orange is the plant's own per-wheel force telemetry, which scores the result and is never admitted into the loop.

58 parameters, full-batch Adam [31] at 5×10^{-4} on an MSE objective, in PyTorch [32]. Because the full-scale residual has different structure from the scaled platform's, the architecture is selectable: `baseline`, `wide` (16 hidden units, 114 parameters, weight decay), `physics_inputs` (74 parameters), `ensemble` (three averaged members, 174 parameters), and `s4` (102–110 parameters). An optional physics-informed loss [33], [13]—dynamics consistency, left/right symmetry and input-gradient smoothness—is available as an ablation switch; the results below use the baseline MSE objective.

1) Physics-augmented inputs: The baseline residual reads $[v_x, v_y, \omega, \delta]$. The quantity the residual actually corrects, however, enters the plant through the slip angles of (3), which are a nonlinear function of all four inputs, so the network has to spend capacity reconstructing them. Appending α_f, α_r computed from (3) with the *measured* geometry gives the network the two arguments of (4) directly, at a cost of 16 parameters and no new sensor. This is the same motivation as the physics-informed regularisers of [33], [13] applied at the input rather than at the loss, and it is available both on its own (`physics_inputs`) and combined with the temporal residual (**s4** with `use_physics_inputs`). The augmentation is rank-agnostic, so the same code path serves a flat batch and a windowed sequence batch.

2) Temporal residual: The `s4` architecture is an S4D diagonal state-space residual [15] over a sliding window of $W = 20$ samples (0.4 s, matched to tire-relaxation timescales). Windows are cut so that none straddles the seam between the recorded trace and its mirrored copy. Each block carries $H = 4$ independent single-input single-output channels of complex state dimension $N = 4$, initialised in the S4D-Lin form ($\mathrm{Re}(A_n) = -\frac{1}{2}$, $\mathrm{Im}(A_n) = \pi n$) that approximates the HiPPO-LegS operator [34] in closed form, with a learnable per-channel discretisation step. As stated in Section II, the S4-family residual overlaps with concurrent work [14] and is not claimed as a contribution; Section V-A3 reports what we did change, which is its cost. Figure 4 shows stage 1 in this configuration.

3) Evaluating the temporal residual at CPU cost: A temporal residual is only usable on-vehicle if it fits inside the re-identification period on the hardware the car carries, which is typically a CPU. Profiled as first written, the S4D path cost 125.7 ms per epoch on a GPU and 898 ms per stage-2 rollout, putting a six-iteration identification at 181.4 s—longer than the 30 s data window it consumes. Three changes remove that cost without altering the arithmetic. None is novel as a technique; what matters here is that the resulting pipeline is real-time-plausible on commodity hardware, and that the equivalence was verified rather than assumed.

a) Closed-form kernel instead of a scan: A diagonal A makes the block's impulse response available in closed form,

$$K_l = \mathrm{Re}\big(C\, e^{A \Delta t\, l}\, \bar{B}\big), \qquad l = 0, \ldots, W-1, \tag{8}$$

so a length-W window is one Vandermonde product followed by a causal Toeplitz matrix product, rather than W sequential steps [15]. At $W = 20$ the dense Toeplitz product is cheaper than the FFT convolution of the original S4 [16] and than a parallel associative scan [20]; both are the right choice at sequence lengths orders of magnitude longer than this one. Because the stack's output layer reads only the window's last timestep, that layer contracts the reversed kernel straight onto the window and skips $W - 1$ of its outputs. Cost: 125.7 ms to 26.7 ms per epoch.

b) Loop-invariant physics-loss terms: Under the physics-informed objective, the nominal Pacejka rollout, the mirrored batch and the constant terms of the symmetry residual depend on the nominal model only, which is fixed for the whole training call, so they are precomputed once per call instead of once per epoch. The three forward passes the objective used to make (data, mirror, smoothness) collapse into one pass over a double-length batch, with the smoothness Jacobian taken off the same graph. Cost: 26.7 ms to 16.6 ms per epoch, with the loss value unchanged bit for bit.

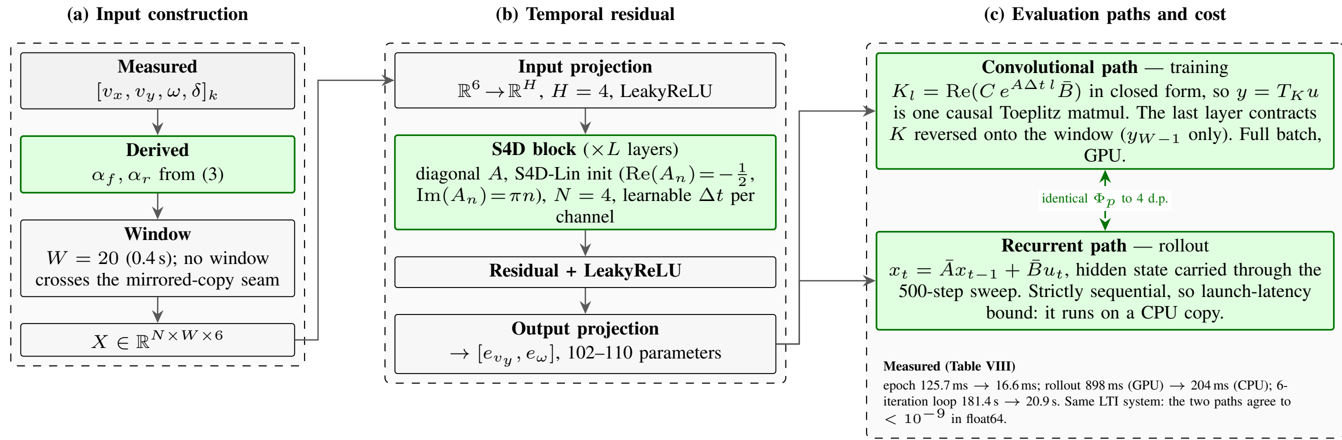

Fig. 4. Stage 1 of Fig. 3 with the physics-augmented input vector (Section V-A1) and the S4D temporal residual (Section V-A2). Green marks what differs from the baseline MLP residual. (a) The four measured signals are augmented with the two slip angles of (3) and cut into $W = 20$ sample windows. (b) The residual stack: input projection to H channels, S4D blocks with a diagonal state-transition matrix, and an output projection to $[e_{v_y}, e_\omega]$. (c) The same linear time-invariant system is evaluated two ways—as a causal convolution with the closed-form impulse response for training, and as a recurrence for stage 2's stateful rollout—which is what makes the architecture affordable on a CPU without changing a single identified coefficient (Section V-A3).

c) The rollout belongs on a CPU: Stage 2 is 500 strictly sequential single-sample steps carrying the S4D hidden state, each with a host synchronisation. That is pure kernel-launch latency on a GPU: 898 ms on CUDA against 204 ms on a CPU copy of the same weights. The rollout is therefore run on a CPU copy whatever device training used—the recurrent path of Fig. 4(c)—and only the full-batch training stays on the GPU where it is 2.4× faster than a CPU. This is the one place where the two evaluation paths of the same system have to agree exactly, and they are pinned by a regression test to $< 10^{-9}$ in double precision.

End to end this takes a six-iteration identification from 181.4 s to 20.9 s (8.7×) on the same synthetic 30 s lap, with the identified Pacejka coefficients agreeing to four decimal places (Table VIII). We deliberately did *not* touch the knobs that trade accuracy for speed—epoch count, iteration count, early stopping, warm-starting the network from the previous iteration's weights, or dropping the smoothness term—because those change the identified tire and are a modelling decision, not a free speedup.

B. Excitation-aware virtual rollout

This is the central correction. Consider stage 2 of Section III-D: a constant-speed rollout at v with the steering ramped to δ_{sweep}. In quasi-steady cornering the lateral acceleration a single-track vehicle develops kinematically is $a_y = v^2 \kappa$ with $\kappa \approx \delta / L$, so the largest friction coefficient the rollout can *demand* of the tire is

$$\boxed{\mu_{\text{reach}} \approx \frac{v^2\, \delta_{\text{sweep}}}{g\, L}} \tag{9}$$

Equation (9) contains no tire parameter: it is a property of the rollout configuration alone. If $\mu_{\text{reach}} < D$, the synthetic data never leaves the linear ramp of (4), only the product BCD is identifiable (Section III-C), and the bounded optimiser resolves the residual freedom on a box edge—precisely the signature reported in Section VI-B.

The baseline implementation selects the rollout speed as $v = \mathrm{clip}(\overline{v_x}, 2, 4)$ m/s, a value representative of a 1:10 platform. On our vehicle, with $\delta_{\text{sweep}} = 0.4$ rad and $L = 1.5334$ m, (9) gives $\mu_{\text{reach}} = 0.43$ at 4 m/s: the rollout cannot ask the tire for more than $0.43\, g$ whatever the tire is. At 1:10 scale, with $L \approx 0.32$ m, the same rollout speed yields $\mu_{\text{reach}} \approx 2.0$ and the pathology is invisible.

1) Correction: The rollout speed is now taken from the speed the vehicle was actually driven at, clamped by configuration bounds (lower bound 7 m/s, no upper bound), and the pipeline warns whenever (9) evaluates below the prior's D—turning a silent degeneracy into a startup diagnostic. The same bound is the right tool for choosing a data-collection speed: identification data should be gathered at $v \geq \sqrt{gLD/\delta_{\text{sweep}}}$.

C. Bounded extrapolation and frozen-prior regularisation

Raising the rollout speed exposes a second, opposite failure. The rollout queries the residual network at steering angles it has never seen: on a pure-pursuit lap the observed $|\delta|$ stays below ≈ 0.06 rad while the sweep ran to 0.4 rad. The fit then reads the network's extrapolation as additional grip, and because each iteration's answer became both the next iteration's nominal model *and* its regularisation target, the error compounds: D walked from the 1.009/1.002 prior to 1.5402/1.8721 over six iterations, a claimed $1.7\, g$ on a vehicle measuring $1.0\, g$ in steady state.

Two changes close the loop:

1) **Extrapolation bound.** The sweep terminates at $\delta_{\text{sweep}} = \mathrm{clip}\big(\lambda_e \cdot P_{99}(|\delta_{\text{obs}}|),\, \delta_{\min},\, \delta_{\max}\big)$ with $\lambda_e = 1.5$, $\delta_{\min} = 0.05$ rad, $\delta_{\max} = 0.34$ rad, so the network is never queried more than 50 % beyond the steering range it was trained on.
2) **Frozen prior.** The Tikhonov target is frozen at the configured prior Φ_p^0 for all iterations, while the optimiser's *start point* continues to track the previous iteration. The

per-axle objective is

$$\min_{\phi} \sum_k \rho\big(r_k(\phi)\big) \;+\; \sum_j w_j\big(\phi_j - \phi_j^0\big)^2, \qquad (10)$$

with r_k the force residual, ρ a soft-ℓ_1 robust loss and w_j scaled to the data residual budget by a single dimensionless weight $\lambda = 0.05$. The regulariser exists specifically to hold the coefficients the manoeuvre does not excite—E above all—at their nominal value instead of on a box edge.

D. Pacejka fit

Fitting applies the steady-state force recovery of (7) to the synthetic data under box constraints $\Phi_p \in ([4.0, 1.2, 0.4, -3.0], [20.0, 2.2, 2.0, 1.0])$ per axle. Relative to the baseline we (i) raised the number of jittered multi-starts from 1 to 8, (ii) added a derivative-free simplex [35] and a global differential-evolution solver [36] as alternatives to trust-region reflective [37], all through the same optimisation library [38], and (iii) corrected the configured prior, which previously initialised D *below* its own lower bound and thus started every optimisation on the rail it then failed to leave. The prior now carries the measured tire of Table **??**.

None of (i)–(iii) fixes the degeneracy on its own, and they were verified as secondary: with the rollout speed left at 4 m/s, a synthetic $D = 1.5$ tire fits back as $D_f = 0.753$, $D_r = 0.737$ with three coefficients railed regardless of solver or start point; at 8–20 m/s the same tire fits back as $D = 1.40$–1.61 with nothing railed.

E. Measurement-side corrections

Three corrections apply to the measurement chain rather than to the estimator.

1) Achieved versus commanded steering: α_f in (3) was built from the `/drive` setpoint. The measured achieved/commanded ratio is 0.76, so α_f was systematically $\approx 25\,\%$ too large and the fitted front stiffness $\approx 20\,\%$ too small (13725 N/rad against the simulator's 17198 N/rad on the same run). Using `feedback/steering_angle` recovers 18486 N/rad.

2) Mass and geometry: m and l_{wb} are taken from the measured static wheel loads (Table I) rather than from the configuration file, and $l_{wb} \equiv l_f + l_r$ is enforced by a regression test, because (7) uses the wheelbase for the normal loads and the axle split for the force division; the two disagreeing is a silent bias.

3) Friction warm-start as a floor: An initial estimate of D is derived from measured accelerations, in the spirit of model-free friction estimation [17]. The baseline-adjacent implementation computed the *median* of $\sqrt{a_x^2 + a_y^2}/g$ over an explicitly *low-slip* mask, which is structurally wrong: it measures utilised friction over samples selected for being far from the limit, and utilisation lower-bounds capability. On a 13 m/s pure-pursuit lap the median measures 0.044 against a peak of 0.46. The estimator is now a 99th percentile over the whole buffer, applied as a *floor* on the prior's D and never as a replacement. Section V-F replaces the bound itself with a tire-model fit. This is a corrected implementation detail rather than a criticism of the friction-estimation literature, whose methods target *available* friction and assume near-limit excitation.

F. Warm-starting the peak factor from measured friction

The 99th-percentile utilisation above is a hard lower bound on the available grip and nothing more: $\sqrt{a_x^2 + a_y^2}/g$ is the friction the vehicle *used*, and it equals the friction the vehicle *had* only at the limit. Under MPC on the raceline this vehicle peaks at 0.63 g against a measured axle μ of 1.05, so the bound is short by $\approx 40\,\%$ by construction, and no choice of quantile fixes that. The warm start is therefore computed from the shape of the axle force curve instead, using only the sensors already on the vehicle.

1) Axle forces and slip angles without a force sensor: Newton–Euler on the single-track body of (1)–(2), solved for the axle forces rather than for the accelerations, gives

$$F_{y,f} = \frac{I_z\dot{\omega} + l_r m a_y}{L\cos\delta}, \qquad F_{y,r} = \frac{l_f m a_y - I_z\dot{\omega}}{L}, \qquad (11)$$

which requires the IMU's lateral acceleration and the yaw acceleration and no tire model at all. Vertical loads carry the longitudinal transfer $F_{z,f/r} = m(g\,l_{r/f} \mp a_x h_{cg})/L$ [28]. The yaw acceleration is taken as the derivative of a local polynomial fit to the yaw rate (Savitzky–Golay, window 21, order 2) [39] rather than by differencing, because $I_z\dot{\omega}$ is a $\approx 10\,\%$ term in $F_{y,f}$ on this vehicle and a differenced odometry yaw rate puts its noise straight into it. Slip angles come from (3) on the buffer that is already collected. Where no IMU is available the accelerations are differentiated from the states by the same filter, and where one is available the RMS difference between the two is logged as a bias diagnostic.

2) Joint fit of stiffness and peak friction: Per axle, both the cornering stiffness and the peak friction are fitted to the Fiala brush model [18], [40], [4]

$$F_y = \mu F_z\big(1 - (1-\xi)^3\big)\operatorname{sgn}\alpha, \quad \xi = \min\Big(\frac{C_\alpha|\tan\alpha|}{3\mu F_z}, 1\Big), \qquad (12)$$

by bounded nonlinear least squares with a soft-ℓ_1 loss [38], [37]. The brush model is used here in place of (4) for one reason: it has two parameters, both of which the manoeuvre can support, whereas (4) has four and is exactly the model whose non-identifiability at low slip this paper is about.

What makes this beat the utilisation bound is that μ enters (12) through the *curvature* of $F_y(\alpha)$ and not only through its asymptote. Writing $r = |F_y|/(C_\alpha|\tan\alpha|)$ for the force's departure from its own linear extrapolation, (12) inverts in closed form to

$$\mu = \frac{2k}{3 - \sqrt{3(4r-1)}}, \qquad k = \frac{C_\alpha|\tan\alpha|}{3F_z}, \qquad (13)$$

so a 30 % departure from linearity ($r = 0.7$) already pins μ, well below the limit. Equation (13) is used only for the excitation diagnostics; C_α and μ are fitted jointly, because a two-stage fit would inherit the bias of estimating C_α on data that is itself already slightly nonlinear.

3) The gate is the feature: Equation (13) also states its own failure mode: $\mathrm{d}\mu/\mathrm{d}r$ grows without bound as $r \to 1$, i.e. as the tire curve stays linear. The estimate is therefore released only when three conditions hold—peak grip utilisation $\max|F_y|/(\mu F_z) \geq 0.40$, relative precision $\sigma_\mu/\mu \leq 0.15$ from the Gauss–Newton covariance, and μ not resting on its own search bound—and otherwise the module logs which condition failed and falls back to the utilisation bound. The 40 % floor is not tuned here: it is the excitation level the lateral-dynamics friction-estimation literature converges on, as surveyed by Acosta *et al.* [19], whose review reports estimators failing below $|a_y| = 0.4\,g$ "due to the proximity of the tyre force curves in the low slip regions" and a 40–80 % utilisation working range. Samples with $|a_x| > 0.35\,g$ are dropped rather than corrected, since the axle longitudinal force split a friction-ellipse correction would need is not observable from this sensor set, and samples whose force opposes their own slip angle are dropped as sign errors or noise.

4) Applied as a floor, unconditionally: The per-axle μ replaces the scalar bound as the initial guess for D_f and D_r where it is released, but—for either input, and however confident the fit—it can only ever *raise* D, never lower it. This is not conservatism for its own sake; two mechanisms depend on it. First, the frozen Tikhonov target of (10) is taken from the warm-started prior immediately after the warm start is applied, so whatever lands there is both the initial guess and the regularisation target of all six iterations. Second, the controller re-ingests the reference speed profile at $\min(a_{y,\max}^{\mathrm{cfg}},\, a_y^{\max}\eta)$ on every accepted model update, so a lowered D rewrites the reference under the running controller; a lowered D that also *moves* between cycles rewrites it repeatedly. Shipping the estimator with a confident fit permitted to replace the prior did exactly that, and the observed behaviour was erratic (Section VI-F1).

The gate cannot substitute for the floor, and it is worth being precise about why: σ_μ/μ from the fit covariance is a *precision* figure, not an accuracy one. A wrong mass, a wrong yaw inertia, an IMU bias or a road bank all fit tightly and wrongly through (11), since axle force scales linearly with m. On this vehicle the configuration file's 240 kg against the simulated 240.0 kg biases μ 12 % low on its own, which is measurement-chain work (Section V-E) and not something a confidence interval can see.

G. Deployment

The identifier runs as a single ROS 2 node at 20 Hz with a rolling FIFO buffer of the collected signals, cycling through four states: collect a full buffer, train, await the controller's acknowledgement of the submitted coefficients, then run—re-identifying every $T_{\mathrm{reid}} = 30\,\mathrm{s}$ from the newest buffer. Three interface details matter for reproducing the results. The achieved road-wheel angle is used when the platform publishes one and falls back to the commanded setpoint on staleness, with the substitution logged (Section V-E). The friction warm start is recomputed on *every* cycle rather than once at startup, which is safe only because the static prior it floors is re-read from configuration each cycle and therefore cannot compound across cycles. Finally, the identified coefficients are submitted over a service call and the node keeps the previous model if the call is not acknowledged, so a rejected identification degrades to the last accepted model rather than to no model; the peak-friction outputs are published on a latched diagnostic topic so a late-starting consumer still sees the current value.

H. Validation protocol

Three levels of validation are used, in increasing strength:

1) **One-step prediction** on a held-out run, as in the baseline: RMSE and R^2 of v_y and ω under $\hat{\mathbf{x}}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k, \Phi_p)$. A dedicated ROS 2 node extends this to configurable multi-step horizons (1/5/10 steps) with Euler/Heun/RK4 integration and online statistics in Welford's form [41].
2) **Direct force comparison** against `feedback/tire_forces`: the identified curve is evaluated at the measured (α, F_z) and compared per axle against the measured F_y. This level is unavailable on a physical platform without a force-measuring hub and is the strongest evidence in the paper.
3) **Closed-loop tracking** under the MPC using the identified model: lap time, mean and maximum lateral deviation, steering sign reversals per control step, and RMS steering rate against the rate limit. This level is *not* completed here; see Section VI-J.

The steady-state force recovery of (7) was itself validated against the simulator's axle force on a normal lap (correlation 0.993, slope 0.94). This matters because (7) is not a force measurement—it is fully determined by $v_x\omega$—so its adequacy had to be established rather than assumed.

VI. Experiments and Results

A. Setup

All experiments run in simulation on the architecture of Section IV: the Formula AI vehicle in CARLA [8], driven through on a ROS 2 Humble , with the CARLA server in synchronous mode at a 0.033 s fixed physics step. Identification data is collected over 30 s of pure-pursuit driving at 50 Hz, matching the baseline's data budget [6]. Ground truth for forces, slip angles and normal loads is the simulator's own per-wheel telemetry.

1) Reference trajectory: Every run follows the same optimised raceline (Fig. 5), produced with the TUM global race trajectory optimization [42] under its minimum-curvature formulation [27] on a recorded centreline of the CARLA silverstone circuit: a closed 5830 m lap of 2916 waypoints at 2.0 m spacing, reference speed 14.90–46.02 m/s, minimum corner radius 18.7 m ($|\kappa|_{\max} = 0.0536\,/\mathrm{m}$).

Two of the optimiser's inputs matter for Section VI-J. Its vehicle configuration of a mass 240 kg, and its `ggv` envelope declares a *constant* lateral capability of 5.0 m/s^2 at every speed. The resulting profile saturates that envelope exactly—peak demand $\max(v_x^2|\kappa|)$ over the lap is 5.00 m/s^2, i.e. 0.56 g—so the trajectory is feasible *by construction* for the vehicle the optimiser was told about, and only for that vehicle.

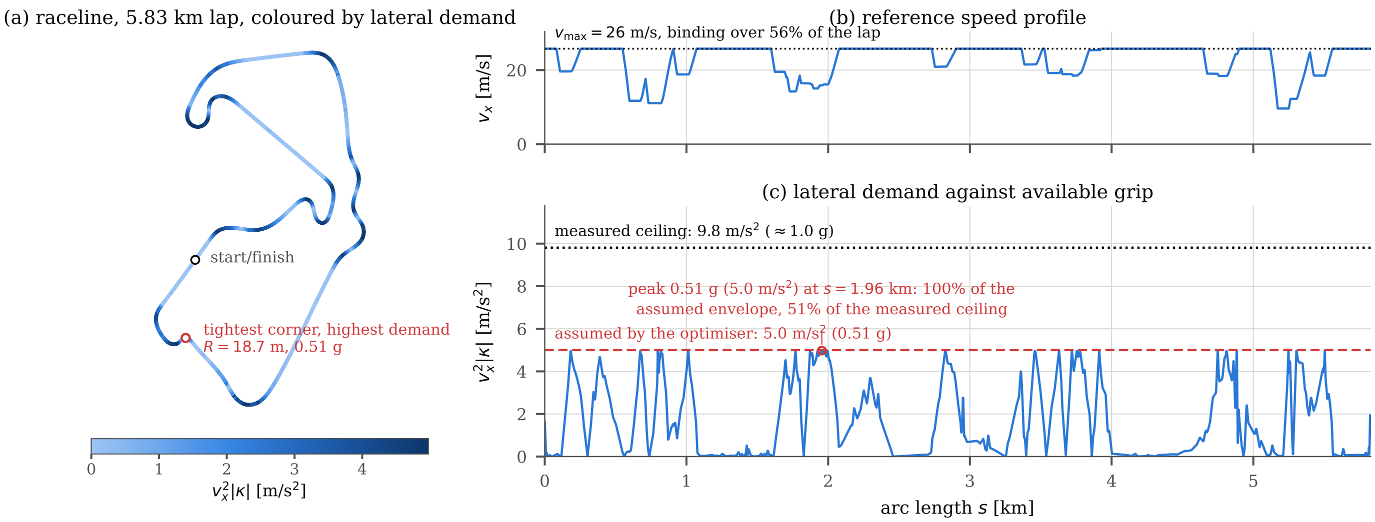

Fig. 5. The optimised raceline used for every experiment, generated with the TUM global race trajectory optimization package [42] under its minimum-curvature formulation [27] from a recorded centreline of the CARLA silverstone circuit. (a) Track layout, coloured by reference speed; the marked corner ($R = 18.7$ m) sets the feasibility limit. (b) Reference speed profile, 14.90–46.02 m/s. (c) Lateral demand $v_x^2|\kappa|$ along the lap. The profile saturates the constant 12.0 m/s^2 envelope given to the optimiser (dashed) but not the grip this vehicle has (dotted, $\approx 1.0\,g$, measured): the shaded segments, 11.9 % of the lap, demand more than the plant can deliver. This is the trajectory the corrected identifier's admissibility gate rejects.

TABLE III
CLOSED-LOOP CONSEQUENCE OF THE DEGENERATE TIRE SET, MEASURED IN A HARNESS DRIVING THE RACELINE FOR 20 S WITH THE PURE-PURSUIT→MPC SWITCH AT $t = 5$ S.

Controller model	Max cross-track	Infeasible stages
Plausible prior ($D = 1.5$)	0.03 m	0
Identified set @ 11.1 m/s	0.16 m	0
Identified set @ 15 m/s	0.43 m	0
Identified set @ 27 m/s	41.89 m	29 461

B. The failure mode, reproduced

Running the baseline pipeline unmodified on this platform yields

$$\text{C\_Pf} = [5.2874,\ 1.5536,\ \mathbf{0.4},\ \mathbf{1.0}],$$
$$\text{C\_Pr} = [19.3236,\ \mathbf{1.2},\ 0.4046,\ -0.1292],$$

with the bold entries exactly on their box constraints and B_r just under its upper bound: five of eight coefficients on or against a rail. The implied physics is qualitatively wrong, not merely inaccurate—a grip ceiling of 3.95 m/s^2 (0.40 g) and a front axle stiffness of 4012 N/rad, so the model believes it needs $\approx$ 0.30 rad of front slip to generate 1 g, beyond the vehicle's 0.2793 rad steering limit. The model believes the car cannot corner. Table III shows why this stays invisible until it is not: lateral demand scales as v^2.

C. Isolating the cause

Four hypotheses were tested against measurement rather than argued from code. Table IV records the outcome, including the two that were wrong; the eliminated hypotheses are as informative as the confirmed one.

The decisive experiment is an offline round-trip with the residual network zeroed, so that only the rollout is under test: a synthetic $D = 1.5$ tire, rolled out at 4 m/s and re-fitted, returns

TABLE IV
HYPOTHESES TESTED AGAINST THE SIMULATOR'S OWN TELEMETRY.

Hypothesis	Verdict
Friction warm-start measures utilised friction over a low-slip mask	*Structurally confirmed, not the cause.* Utilisation measures 0.044 median, 0.46 peak at 13 m/s. Corrected anyway (Sec. V-E).
Static D prior below its own lower bound	*Confirmed, secondary.* Corrected; multi-starts 1→8.
Only BCD identifiable at low slip	*Confirmed, and it is the mechanism*—but caused by the rollout speed, not by the recorded lap (Sec. V-B).
`/odom` twist in world frame	*Refuted.* corr $= 0.93$, RMS 0.09 m/s against a body-frame interpretation.
Steady-state force model (7) inadequate	*Refuted as a cause.* corr $= 0.993$, slope 0.94 against the simulator's own axle force.
δ taken from command, not achieved	*Confirmed, significant.* Ratio 0.76; worth 20 % of front stiffness.

$D_f = 0.753$, $D_r = 0.737$ with three coefficients railed; the same tire rolled out at 8–20 m/s returns $D = 1.40$–1.61 with nothing railed. Since the tire is known exactly and the residual model is absent, this isolates the rollout configuration as the cause and confirms (9). Figure 24 shows the round trip as a sweep over rollout speed, next to the reachable-friction bound that predicts it and the slip-angle coverage each configuration delivers to the fit.

D. Baseline versus this work

Table V places the pipeline as published [6] beside the pipeline as it stands here, on the axes that changed. The residual model, the data budget, the four-stage loop and the steady-state force recovery are inherited unchanged; what differs is the configuration of the virtual-data step, the regularisation topology, the measurement chain, and the contract at the output.

TABLE V
THE BASELINE PIPELINE [6] AGAINST THIS WORK, ON THE AXES THAT DIFFER. ROWS MARKED *inherited* ARE UNCHANGED AND ARE LISTED SO THAT THE EXTENT OF THE MODIFICATION IS UNAMBIGUOUS.

Axis	Baseline [6]	This work
Platform	1:10, 3.7 kg, $L = 0.32$ m	Full scale, 240.0 kg, $L = 1.53$ m
Rollout speed	$\mathrm{clip}(\overline{v_x}, 2, 4)$ m/s	From the measured lap, floored at 7 m/s
Reachable μ	≈ 2.0 (adequate)	0.43 under the baseline clip; ≥ 1.2 after correction
Sweep amplitude	Fixed 0.4 rad	$1.5P_{99}(\lvert\delta_{\mathrm{obs}}\rvert)$, clipped to $[\delta_{\min}, \delta_{\max}]$
Regularisation	None; each fit is the next start point	Tikhonov to a *frozen* prior; start point still tracks
Steering signal	Command	Achieved road-wheel angle
Mass, geometry	Configuration file	Measured static wheel loads
Friction prior	—	Per-axle brush fit of (C_α, μ) from IMU and odometry, gated; P_{99} of $\lvert a\rvert/g$ as fall-back; always a floor
Solver	Trust-region, single start	Multi-start, plus simplex and global options
Output contract	Per-coefficient box constraints	Gates on derived quantities at the controller interface
Residual model	MLP, 58 parameters	*inherited* as default; selectable wide, physics-augmented, ensemble and S4D temporal alternatives
Residual cost	Not reported	S4D identification 181.4 s $\rightarrow$ 20.9 s, exact (Table VIII)
Data budget	30 s, one controller-free lap	*inherited*
Force recovery	Quasi-steady yaw balance	*inherited*
Ground truth	Steady-state manoeuvre in a large space	Simulator per-wheel force, slip and load

E. Identification after correction

With the six defects of Table VI corrected, three consecutive live identifications return no coefficient on a bound. Peak friction coefficients land at $D_f, D_r \in [1.05, 1.18]$ against the reference 1.009/1.002 of Table **??**; axle cornering stiffnesses land at $C_f \approx 21$ kN/rad and $C_r \approx 20$ kN/rad against the simulator's measured 17–19 kN/rad; the understeer gradient has the correct sign ($l_f C_f < l_r C_r$) on every run, matching the measured plant; and one-step prediction on held-out data gives $R^2 = 0.97$ for v_y and $R^2 = 0.99$ for ω. The peak-friction drift across the six co-identification iterations is removed: where the unbounded sweep walked D from 1.01 to 1.87, the bounded sweep with a frozen prior settles inside $[1.05, 1.18]$.

F. Peak-friction warm start

The friction warm start of Section V-F is evaluated on a synthetic single-track plant with brush tires rather than in CARLA first, because only a synthetic plant supplies a μ that is known exactly and an excitation level that can be swept independently of it. The plant is driven by a swept-sine steering input for 40 s at 50 Hz with true $\mu = 1.05$ and $C_\alpha = 12.1F_z$, and the steering amplitude sets how far up the tire curve the manoeuvre reaches. Table VII reports what each estimator returns.

At 45 % utilisation—a level an ordinary lap reaches—the utilisation bound is 55 % low while the brush fit is within 0.2 %. Across true $\mu \in \{0.7, 1.05, 1.4\}$ the fit stays within 5 % and C_α within 10 %; with odometry and IMU noise injected, μ stays within 12 %. The gate behaves as designed: below the 40 % floor it suppresses the fit and reports the bound, so the estimator's failure is visible rather than silent. Figures 15–17 are the corresponding diagnostics on the CARLA plant and are not yet collected.

1) Negative result: a confident fit must still not lower D: The estimator was first shipped with the natural policy—replace the prior's D when the gate passes, in either direction—and the vehicle became erratic within one session. The mechanism is the one anticipated in Section V-F4: a μ estimate slightly below the prior lowers D, the lowered D becomes the frozen regularisation target of every subsequent iteration, and the controller re-derives the reference speed profile from the identified grip ceiling on each accepted update, so corner speeds collapse and the high-curvature stages lose their steady-state equilibrium. Because the warm start is recomputed every cycle, a wandering estimate rewrote that reference repeatedly. The confidence gate could not have caught it, for the reason given in Section V-F4: σ_μ/μ is precision, not accuracy. The policy is now floor-only for every input kind, and we report the episode because the failure is a property of the identification-to-control interface rather than of the estimator, and any stack that lets an online estimate move a reference trajectory inherits it.

G. Identification cost

Section V-A3's claim is that the temporal residual is affordable on the hardware a vehicle carries; Table VIII is the measurement, on the same synthetic 30 s lap and the same six co-identification iterations throughout. Every step was verified against the implementation it replaced before being kept: the convolutional and recurrent paths of Fig. 4(c) agree to $< 10^{-9}$ in double precision (pinned by a regression test), the rewritten physics-informed loss is bit-identical, and the full six-iteration loop lands on the same Pacejka coefficients to four decimal places—the remaining difference is float32 reordering over 1200 optimiser steps.

Two consequences are worth stating. First, at 20.9 s the identification fits inside the 30 s re-identification period of Section V-G, so the temporal residual is a deployable option rather than an offline one; at 181.4 s it was not. Second, the device split is not incidental: the stage-2 rollout is strictly sequential and single-sample, so it is 4.4$\times$ *faster* on a CPU, while full-batch training over ≈ 3000 windows remains 2.4$\times$ faster on the GPU. A pipeline that pins itself to one device pays one of those two penalties.

H. Ablations

Three ablation axes are exposed by the implementation and are marked as not yet run rather than reported with placeholder numbers. The first is the residual architecture (`baseline` MLP, `wide`, `physics_inputs`, `ensemble`, `s4`), scored on one-step and 10-step RMSE and on the identified coefficients (Fig. 21); the relevant comparison is [14], which reports a lateral-force RMSE reduction exceeding 60 % for an S4 residual, and we expect a smaller effect here because our sequences are short and our residual is dominated by

TABLE VI
DEFECTS IDENTIFIED, CORRECTED AND REGRESSION-TESTED IN THIS WORK. THE EFFECT COLUMN REPORTS THE MEASURED CHANGE WHERE ONE WAS QUANTIFIED.

Defect	Root cause	Correction	Measured effect
Degenerate fit, D on its bound	Pacejka fit runs on a synthetic rollout whose speed was clipped to 2–4 m/s; $\mu_{\text{reach}} = 0.43$	Rollout speed from measured lap speed, floored at 7 m/s; warning when (9) $< D$	D: 0.40 (railed) $\rightarrow$ 1.05–1.18; C_f: 4.0 $\rightarrow$ 21 kN/rad
Identified set never reached the controller	Trainer wrote coefficients to file but returned nothing; the node re-read a static prior file and submitted *that*	Trainer returns the identified coefficients; node submits them	Service received the prior verbatim while the fit had produced a different set
D drifts upward across iterations	Sweep extrapolated the residual network far past its training range, and each answer became its own regularisation target	Sweep bounded at $1.5 \times P_{99}(\lvert\delta\rvert)$; regularisation target frozen at the configured prior	1.01 $\rightarrow$ 1.87 drift removed; settles 1.05–1.18
Front stiffness underestimated	Slip angle built from the steering *setpoint*	Achieved road-wheel angle from feedback/steering angle, with fallback	C_f: 13725 $\rightarrow$ 18486 N/rad (truth 17198)
Friction prior pinned near 0.4	Median of *utilised* friction over an explicitly *low-slip* mask; utilisation lower-bounds capability at any quantile	P_{99} over the whole buffer as a floor, then a gated per-axle brush fit of (C_α, μ) (Sec. V-F)	Median 0.044 vs peak 0.46 at 13 m/s; brush fit recovers $\mu = 1.048$ against truth 1.05 at 45 % utilisation, where the bound reads 0.473 (Table VII)

TABLE VII
PEAK FRICTION RECOVERED FROM A SYNTHETIC BRUSH-TIRE PLANT (TRUE $\mu = 1.05$) AS A FUNCTION OF EXCITATION. THE UTILISATION ESTIMATOR TRACKS THE GRIP THE MANOEUVRE *used*; THE BRUSH FIT OF (12) RECOVERS THE PEAK FROM THE CURVATURE OF $F_y(\alpha)$ AND IS THEREFORE FLAT IN EXCITATION. BELOW 40 % UTILISATION THE BRUSH FIT IS GATED OFF AND THE UTILISATION BOUND IS REPORTED INSTEAD.

δ amplitude	Grip utilisation	Utilisation	Brush fit (12)
0.05 rad	45 %	0.473	**1.048**
0.07 rad	63 %	0.658	**1.048**
0.09 rad	80 %	0.834	**1.048**
0.13 rad	100 %	1.045	**1.048**

TABLE VIII
COST OF ONE S4D IDENTIFICATION, MEASURED END TO END ON A SYNTHETIC 30 S LAP. THE OPTIMISATIONS ARE EXACT: NO EPOCH COUNT, ITERATION COUNT, EARLY-STOPPING RULE OR LOSS WEIGHT WAS CHANGED, AND THE IDENTIFIED COEFFICIENTS AGREE TO FOUR DECIMAL PLACES.

Stage	Device	Before	After
Training epoch: scan $\rightarrow$ kernel (8)	GPU	125.7 ms	26.7 ms
Training epoch: physics-loss hoist and fusion	GPU	26.7 ms	16.6 ms
Stage-2 rollout, 500 sequential steps	GPU $\rightarrow$ CPU	898 ms	204 ms
Identification, 6 iterations	mixed	181.4 s	20.9 s

a correctable measurement bias rather than by unmodelled temporal dynamics. The cost objection to running this axis is now removed (Section VI-G): the S4D arm costs 20.9 s per identification rather than 181.4 s. The second is the loss mode (`mse` versus `physics_informed`) on the same dataset, comparing one-step metrics, per-iteration coefficient trajectories and closed-loop tracking. The third is the input vector: whether the slip-angle augmentation of Section V-A1 buys anything once the residual is temporal, which is a question about whether α_f, α_r are recoverable from a 0.4 s window (Fig. 22).

I. adaptation to time-varying friction

The experiments reported above hold the surface fixed and ask whether the tire model is recovered correctly. The complementary question—whether the identifier tracks a surface that *changes* while the vehicle is on it—is evaluated with the three friction schedules introduced by LLA-MPC [43], reproduced here on the CARLA plant so that the comparison is on the same axes as the adaptive-racing literature. Friction is varied by rewriting the per-wheel `tire_friction` of the vehicle's physics control at the simulator's fixed 0.033 s step, so that the schedule is exact and known, while the identifier sees only IMU, odometry and steering feedback as before. In every case the plant's own per-wheel telemetry supplies the true $\mu(t)$ against which the estimate is scored, and the same re-identification period of Section V-G is used throughout.

1) Gradual friction decay (Experiment 1): The actual friction coefficient decays linearly at 2 % per second, as in [43]. This is the benign case: the surface evolves slowly enough that a controller with a stale model degrades gradually rather than failing outright, so it measures tracking latency rather than recovery. The quantity of interest is the lag between $\mu(t)$ and the identified $D(t)$, and whether the floor-only warm-start policy of Section VI-F1 biases the estimate upward when grip is falling monotonically—a case the floor was never designed for, and the one place where the policy is expected to cost accuracy.

2) Sudden friction drop (Experiment 2): A step reduction of 40 % in the friction coefficient at the end of the first lap, again following [43]. Physically this stands for a dry-to-wet transition, an oil or debris patch, or a change of surface material; none of these leave time for gradual adaptation. The step is placed after one completed lap so that every method has a full lap of data before it occurs. What is measured is the recovery time—the interval from the step to the first identification whose D is within 5 % of the new truth—and the closed-loop cost paid in the interim.

Reporting is on the three metrics used in [43]—lap time, maximum cross-track error and a violation count—plus the friction-tracking error $|\hat{\mu}(t)-\mu(t)|$, which is the identifier-level quantity this work is actually about. Figures 19 and 20 are the corresponding diagnostics. All three experiments are closed-loop and therefore inherit the prerequisite of Section VI-J: they are blocked until the raceline is regenerated for the measured vehicle, since a trajectory already infeasible at nominal grip cannot be driven at 60 % of it.

J. Negative result: the raceline, not the model

With the identifier corrected, level 3 of the validation protocol (Section V-H) does *not* complete, and the reason is instructive. The identified model reports a grip ceiling of 1.02–1.16 g; the plant sustains $\approx 1.0\,g$ in steady state; the raceline demands $\max(v_x^2|\kappa|) = 12.00\,\text{m/s}^2 = 1.22\,g$ ($P_{99} = 11.90\,\text{m/s}^2$). The admissibility gate of Section VII therefore rejects every identification, and does so correctly: the trajectory is infeasible for this vehicle, and the corrected identifier is what made that visible.

The cause is worth naming because any model-based stack can inherit it. The trajectory optimiser was handed a $\mathtt{ggv}$ envelope asserting a constant 12.0 m/s² of lateral capability and a vehicle mass of 204 kg, and it did its job—the profile saturates that envelope to the last digit. The envelope was not this vehicle's. Identification and trajectory generation were parameterised from two different descriptions of the same car, and nothing in the stack compared them until the gate did.

We do not loosen the gate. The quantitative consequence for regeneration is that the tightest corner, $|\kappa| = 0.0536\,$/m, permits $\sqrt{g\mu/|\kappa|} = 13.5\,\text{m/s}$ at the measured grip (13.7–14.6 m/s at the identified ceiling of 1.02–1.16 g); capping the speed profile at 13 m/s brings peak demand to 9.05 m/s² (0.92 g), whereas 15 m/s returns it to 12.00 m/s². Regenerating the raceline for the *measured* vehicle ($m = 240.0\,\text{kg}$, $\mu_{\text{peak}} = 1.0$, $v_{\max} \approx$ 13–14 m/s) is the prerequisite for the closed-loop experiment and is the immediate next step (Section IX).

VII. Integration with the Controller

The identified model leaves the pipeline through one interface. A nonlinear MPC path-tracking controller (acados [25], partial-condensing HPIPM [26], horizon $N = 20$ at a 20 Hz control period) carries the same single-track model as the identifier and consumes Φ_p directly: the eight Pacejka coefficients enter the prediction model's axle forces through (4), and nothing else about the controller changes when a new identification arrives. Coefficients are pushed over a ROS 2 service at the end of each identification cycle, so the controller's model tracks the plant at the re-identification rate. We claim no contribution to the controller design; it appears here as the consumer that defines what an admissible identification is.

Six stacked panels sharing a 30 s time axis.

Fig. 6. Example telemetry from one identification run. (a) v_x; (b) v_y and ω; (c) commanded versus achieved road-wheel angle, whose ratio measures 0.76; (d) a_x, a_y; (e) per-axle lateral force from the simulator's wheel telemetry against the value recovered by the steady-state relation (7); (f) per-axle slip angle.

μ_{reach} versus rollout speed, one curve per sweep amplitude, with the old clip band and the plant's measured μ.

Fig. 7. Reachable friction of the synthetic rollout, $\mu_{\text{reach}} = v^2\delta_{\text{sweep}}/(gL)$, for the full-scale vehicle ($L = 1.53$ m, solid) and a 1:10 platform ($L = 0.32$ m, dashed). The shaded band is the rollout speed inherited from the scaled platform; within it the full-scale rollout cannot demand more than $\mu = 0.43$, so the tire's peak factor is not identifiable from it. The same configuration is comfortably above the peak at 1:10, which is why the defect is scale-latent.

A. Admissibility gates

A coefficient set can lie strictly inside every box constraint and still be unusable. We observed three distinct ways, each requiring a gate on a *derived* quantity rather than on a coefficient:

1) **Grip ceiling versus trajectory demand.** The controller computes $\max_k(v_{x,k}^2|\kappa_k|)$ over the speed-clamped reference and rejects any identification whose $a_y^{\max}$ from (6) falls below it. The degenerate fit reported a 0.40 g ceiling against a 1.22 g demand and produced 41.89 m of cross-track error at 27 m/s, against 0.03 m for a physically plausible set (Table III).
2) **Understeer sign and critical speed.** A set with $l_f C_f > l_r C_r$ whose v_{crit} falls inside the controller's speed range makes the prediction model open-loop unstable, and the resulting $\rho(A_d) > 1$ is raised to ρ^N in the condensed QP. An observed set gave $C_f = 37031$ N/rad against $C_r = 11691$ N/rad, i.e. $v_{\text{crit}} = 17.4$ m/s.

(a) Slip-angle histograms before/after; (b) (α, F_y) scatter with fitted curve, before/after.

Fig. 8. Excitation entering the Pacejka fit. (a) Distribution of the slip angles present in the synthetic data before and after the rollout correction, with the reference tire's peak-force slip angle marked. (b) The corresponding (α, F_y) scatter with the fitted curve; before the correction all data lies on the initial linear ramp, where only the product BCD is identifiable.

2×4 grid: front/rear $\times$ B, C, D, E versus co-identification iteration, two series.

Fig. 9. Coefficient trajectories over the six co-identification iterations, with the unbounded sweep and tracking regularisation target (which walks D from 1.01 to 1.87) against the bounded sweep with a frozen target (which settles in $[1.05, 1.18]$). Dashed lines mark the box constraints and the reference values of Table **??**.

3) **Integration stiffness.** Even a stable set can be stiff relative to the control period: a single explicit RK4 step at $dt = 0.05\,\mathrm{s}$ gave $\rho(A_d) = 547$ at $5\,\mathrm{m/s}$ for one identified set. The model sub-steps adaptively, sizing the sub-step from the model's own fastest lateral mode $|\lambda| \approx (C_f + C_r)/(mv_x) + (l_f^2 C_f + l_r^2 C_r)/(I_z v_x)$ so that $|\lambda| h \leq 1.5$, restoring $\rho \approx 1.000$ across the speed range at 2.3–6.8 ms per 100-stage horizon.

The gates are the contract between identification and control: an identification that passes is a model the controller can be held responsible for, and one that fails is reported as unidentified rather than silently deployed. Two further controller-side changes were required before any identified model could be evaluated at full-scale speeds and are recorded for completeness rather than as identification results: reference speeds are clamped at ingest, and the solved state is rolled forward by the measured transport delay before each solve, which took cross-track error at $27\,\mathrm{m/s}$ from $130.55\,\mathrm{m}$ to $0.38\,\mathrm{m}$.

VIII. Discussion and Limitations

A. What generalises

The reachable-friction bound (9) is kinematic: it involves no tire parameter, no simulator-specific quantity and no property

Two panels (front, rear): $F_y(\alpha)$ for reference, degenerate, and corrected fits over the measured scatter.

Fig. 10. Identified tire models against the plant. Each panel shows the reference fit of Table **??**, the degenerate result of the untransplanted pipeline, and the corrected result over three consecutive identifications (shaded band), with the raw per-wheel (α, F_y) measurements underneath.

Measured versus predicted v_y and ω on held-out data, with residual histograms inset.

Fig. 11. One-step prediction on a held-out run using the corrected identified model: $R^2 = 0.97$ for lateral velocity and $R^2 = 0.99$ for yaw rate.

of the residual model, so it applies to any implementation of the virtual-steady-state-data strategy of [6], in simulation or on hardware. Its practical content is a design rule—*the rollout must be configured to demand at least the friction one intends to identify*—and, where it cannot, D should be reported as unidentified rather than as fitted. The diagnostic signature, coefficients resting exactly on box constraints, is likewise generic and should be checked and logged by any bounded tire-fitting routine, since it is otherwise indistinguishable in the output from a successful fit.

The insufficiency of per-coefficient box constraints for deployment admissibility is also platform-independent. Both failures we encountered—a set that satisfies every box constraint while being degenerate, and a set that satisfies every box constraint while being pairwise oversteering and unstable within the controller's operating range—are properties of the parameterisation, not of CARLA. Deep Dynamics [13] constrains coefficients inside the network; we argue that a second, derived-quantity check belongs at the interface where the model is handed to a controller.

B. What does not generalise

Three results are properties of this plant and must not be read as statements about tires. First, the numerical tire values: the plant is a PhysX wheeled-vehicle model with a configured friction coefficient, so Table **??** describes *that* plant, and agreement with it is evidence that the identifier recovers the plant it observes, not evidence about real tire

Pending: demand profile after regenerating the raceline for the measured vehicle, on the same axes as Fig. 5(c).

Fig. 12. Regenerated speed profile. **Not yet produced**: the demand-versus-grip comparison for the *current* trajectory is Fig. 5(c); this figure is its counterpart after the raceline is regenerated for the measured vehicle ($m = 240.0\,\mathrm{kg}$, $\mu_{\mathrm{peak}} = 1.0$), which is the prerequisite for the closed-loop experiment of Fig. 14.

Grouped bars: C_f, C_r across five model sources.

Fig. 13. Axle cornering stiffness $C = F_z BC^{\mathrm{MF}} D$ across model sources: simulator ground truth, the degenerate fit, a fit referenced to the steering command, a fit referenced to the achieved steering angle, and the final corrected fit. The understeer condition $l_f C_f$ versus $l_r C_r$ is annotated per group.

Pending: cross-track error versus time, and driven path over the raceline.

Fig. 14. Closed-loop tracking under MPC with the identified model. **Not yet produced**: this experiment is blocked on regenerating a feasible raceline for the measured vehicle, see Section VI-J.

Pending: two panels (front, rear). (α, F_y) scatter recovered from IMU and odometry, the fitted brush curve, the linear extrapolation $C_\alpha \tan\alpha$, and the identified μF_z against the plant's measured peak.

Fig. 15. The peak-friction warm start on the CARLA plant. **Not yet produced.** Per axle: the lateral force reconstructed by Newton–Euler (11) against the slip angle, the joint (C_α, μ) fit of the brush model (12), and the departure from the linear extrapolation that identifies μ through (13). The plant's own per-wheel force telemetry underlies the scatter as ground truth and is never used by the estimator.

behaviour. Sagmeister *et al.* [22] show that sensitivity to model fidelity is largest near the acceleration limits, which is where this study operates, and R-CARLA [21] exists because CARLA's native dynamics are a known weak point for racing-grade work; hardware replication is therefore necessary before any claim about identification accuracy on a real vehicle. Second, the gap between configured wheel friction (1.5) and realised peak axle friction (1.00–1.05) is a property of how the PhysX solver composes friction with load and slip; we report it only because an identifier tuned to converge on the number in the configuration file will converge on the wrong number. Third, the 0.76 achieved/commanded steering ratio arises from CARLA's Ackermann controller and this vehicle's steering configuration—the lesson, identify against the achieved actuator state, is general and standard practice; the magnitude is not.

C. Limitations

1) **Closed-loop validation is incomplete.** Level 3 of Section V-H has not been run end to end, because the admissibility gate correctly rejects the model for the raceline in hand (Section VI-J). Until a feasible raceline is regenerated this paper reports open-loop accuracy and closed-loop *consequence* (Table III), not closed-loop racing performance. We consider stating this preferable to relaxing the gate.
2) **No noise sweep.** The baseline's robustness claim—a 3.3× RMSE advantage over nonlinear least squares under injected noise [6]—has not been reproduced here. The bridge's odometry noise model exists and is disabled for these runs, so the experiment is available; it has not been run, and our results are therefore not directly comparable to the baseline's robustness figures.
3) **No same-data NLS baseline.** We compare against the plant's own telemetry rather than against a nonlinear least-squares identification of the same data. That comparison would strengthen the paper and is straightforward to add.
4) **Single surface, vehicle and trajectory.** All results come from one vehicle configuration on one map. The runtime friction override supports a surface-change adaptation experiment—arguably the strongest argument for on-track identification—but it is not performed here.
5) **Yaw inertia is assumed, not measured** (Section A). This bears directly on the friction warm start: the axle forces of (11) scale linearly with m and carry $I_z\dot{\omega}$ as a $\approx 10\,\%$ term, so both feed straight into μ without widening its confidence interval (Section V-F4). The estimator additionally assumes a flat road, an IMU at the centre of gravity and pure lateral slip; a road bank leaks $g\sin\phi$ into a_y, a longitudinal IMU offset x adds $\dot{\omega}x$, and combined-slip samples are dropped rather than corrected because the axle longitudinal force split a friction-ellipse

Pending: identified μ versus grip utilisation, three series (truth, utilisation bound, brush fit), with the 40 % gate threshold marked.

Fig. 16. Excitation dependence of the two friction estimators. **Not yet produced**; the synthetic-plant counterpart is Table VII. The utilisation bound rises with the grip the manoeuvre happens to use; the brush fit is flat in excitation above the gate. The shaded region below 40 % utilisation is where the module refuses to answer and reports the bound instead—the boundary reported across the lateral-dynamics friction literature [19].

Pending: (a) σ_μ/μ versus utilisation with the 0.15 gate; (b) IMU a_y against the odometry-differentiated a_y, with the RMS difference annotated.

Fig. 17. Confidence and consistency diagnostics. **Not yet produced.** (a) The relative precision of the fitted μ collapses as the tire curve stays linear, which is what the gate of Section V-F tests. (b) The cross-check between the IMU's lateral acceleration and the same quantity differentiated from odometry: a persistent offset here is an IMU bias or a mounting offset, both of which bias μ through (11) without widening σ_μ (Section V-F4).

correction needs is not observable from this sensor set.

6) **The friction estimator is validated on a synthetic plant, not yet in CARLA** (Table VII). It is a synthetic plant chosen precisely because μ is known there and excitation can be swept independently of it, but the CARLA cross-check against the simulator's own per-wheel forces (Figs. 15–18) is not collected, and the shipped configuration therefore still selects the utilisation bound by default.
7) **Ablations not completed**, so no claim is made about the accuracy of the S4D residual, the augmented inputs or the physics-informed loss on this platform (Section VI-H); in particular we do not claim to reproduce the S4 improvement of [14]. What is claimed about the S4D residual here is its cost and the exactness of the reduction (Table VIII), which is a statement about the implementation and says nothing about whether the architecture identifies a better tire.
8) **Longitudinal dynamics are out of scope.** As in the baseline only lateral dynamics are identified; combined slip and load transfer are neglected, and the simulator's longitudinal wheel force channel is drivetrain torque rather than a contact-patch force, so it is unusable as ground truth for a longitudinal extension without further

Pending: D_f, D_r, warm-start μ and the utilisation bound versus identification cycle over a multi-lap run, with the floor-only and replace policies overlaid.

Fig. 18. Warm start across identification cycles. **Not yet produced.** The identified peak factors and the warm-start inputs published on the diagnostic topic of Section V-G over consecutive cycles. The second series is the reverted policy of Section VI-F1, in which a confident fit was allowed to lower D and the reference speed profile was rewritten under the running controller on every cycle.

Pending: three stacked panels sharing a time axis, one per friction schedule; true $\mu(t)$ against the identified $D(t)$ and the warm-start estimate, with lap boundaries and the change instant marked.

Fig. 19. Friction tracking under the three schedules of Section VI-I. **Not yet produced.** (a) Linear decay at 2 % per second; (b) a 40 % step at the end of the first lap; (c) the same step within the first lap. Each panel shows the simulator's commanded $\mu(t)$ as ground truth, the identified peak factor $D(t)$ published on the diagnostic topic of Section V-G, and the warm-start input that seeds it (Section V-F). The quantities read off are the tracking lag in (a) and the recovery time in (b) and (c); the schedules follow LLA-MPC [43] so that the axes match the adaptive-racing literature.

work.

D. Threats to validity

The most serious internal threat is that several corrections of Table VI were applied together rather than factorially; only the rollout-speed correction was isolated, by the zeroed-residual round-trip of Section VI-C. The effects attributed to individual measurement-side corrections come from re-fitting the same recorded run with one change at a time, which controls for the data but not for interaction between changes; a full ablation would settle this and is inexpensive offline. The most serious external threat is the simulator's tire model, discussed above; the mitigation available within this study is that every claim is scored against the plant's own per-wheel forces rather than against a secondary model, so the identification result is at least not self-referential.

IX. Conclusion and Future Work

This paper studied the identifiability of a learning-based on-track Pacejka identification pipeline [6] at full vehicle

Pending: driven paths over the raceline for the three schedules, coloured by speed, with off-track excursions and gate rejections annotated.

Fig. 20. Closed-loop consequence of a changing surface. **Not yet produced**; blocked on the regenerated raceline of Section VI-J. The driven path under each of the three friction schedules against the reference, coloured by speed, with the instant of the change marked and every admissibility-gate rejection (Section VII) annotated on the lap. Experiment 3 is the adversarial case: the step arrives before a full identification buffer exists, and the gate should reject rather than let the controller pursue a trajectory the surface no longer supports.

Pending: grouped bars, one-step and 10-step RMSE of v_y and ω across the five residual architectures, with identified D_f, D_r annotated per group.

Fig. 21. Residual-architecture ablation. **Not yet produced**; Section VI-H. The five selectable architectures (`baseline`, `wide`, `physics_inputs`, `ensemble`, `s4`) on the same recorded lap, scored on held-out one-step and 10-step prediction and on the coefficients each one leads the Pacejka fit to. The comparison of interest is against the 60 % lateral-force RMSE reduction reported for an S4 residual in [14].

Pending: (a) residual RMSE versus window length W; (b) with and without the slip-angle augmentation, per architecture.

Fig. 22. What the temporal residual and the augmented inputs each buy. **Not yet produced.** (a) Prediction error against the window length of Section V-A2, whose default $W = 20$ (0.4 s) is set by tire-relaxation timescales rather than by a sweep. (b) The slip-angle augmentation of Section V-A1 on the MLP and on the S4D residual: whether handing the network the arguments of (4) still helps once it can see 0.4 s of history.

Pending: stacked runtime bars per identification stage, before and after, split by device; inset showing conv-versus-scan agreement in float32 and float64.

Fig. 23. Where an S4D identification spends its time. **Not yet produced**; the numbers are in Table VIII. The stacked bars break the 181.4 s down by stage and device and show what each of the three changes of Section V-A3 removes. The inset is the equivalence check: the convolutional and recurrent evaluations of the same state-space block agree to $< 10^{-9}$ in double precision, which is what licenses calling the speedup exact.

scale, using a CARLA simulation architecture that exposes the plant's own per-wheel forces as ground truth. Transplanted unchanged from a 1:10 platform, the pipeline fails in a specific, diagnosable and correctable way. The failure is not a solver, bounds or tuning problem: it is a loss of excitation created by the pipeline's own virtual-data step, whose reachable friction $\mu_{\text{reach}} = v^2\delta_{\text{sweep}}/(gL)$ depends on the rollout configuration and not on the tire. A rollout speed appropriate at 1:10 caps μ_{reach} at 0.43 at full scale, below the plant's peak, so the peak factor is not identifiable from the synthetic data and the bounded optimiser resolves the ambiguity on a box edge.

Correcting this—together with an extrapolation bound on the steering sweep, a frozen regularisation target, identification against the achieved rather than the commanded road-wheel angle, measured mass and geometry, and a friction warm start identified from the curvature of the axle force curve and applied only as a floor—produces identifications with no railed coefficient, peak friction coefficients of 1.05–1.18 against a measured 1.00–1.05, axle cornering stiffnesses within 10–20 % of the simulator's own, correct understeer sign, and one-step R^2 of 0.97/0.99. We further argue that per-coefficient box constraints are not a sufficient admissibility criterion for handing a model to a model-based controller, and place gates on derived physical quantities at that interface instead. The corrected identifier's first useful act was to reject the reference trajectory: the raceline in hand demands $1.22\,g$ from a vehicle that sustains $1.0\,g$. We report this rather than relaxing the gate.

A. Future work

a) Immediate:

1) Regenerate the raceline for the *measured* vehicle ($m = 240.0\,\text{kg}$, $\mu_{\text{peak}} = 1.0$, $v_{\max} \approx$ 13–14 m/s) and run the closed-loop validation: lap time, mean and maximum lateral deviation, and steering sign reversals per control step, against a hand-tuned prior model as control.
2) Run the noise sweep of [6] on this platform using the bridge's odometry noise model, with a nonlinear least-squares identification of the same data as the comparison arm.
3) Complete the residual-architecture, input-augmentation and loss-mode ablations of Section VI-H, and a factorial ablation over the six corrections of Table VI. The runtime objection to the temporal arm is removed (Section VI-G).
4) Collect the friction-estimator diagnostics of Figs. 15–18 on the CARLA plant. The estimator is so far validated against a synthetic brush-tire plant, where μ is known

Pending: three panels: (a) μ_{reach} versus rollout speed at full scale and 1:10; (b) round-trip identified D versus rollout speed under the baseline and the excitation-aware configuration; (c) the $(\alpha_f, F_{y,f})$ coverage each configuration hands to the fit.

Fig. 24. The baseline pipeline [6] against this work, with the residual network held at zero so that the rollout configuration is the only variable. **Not yet produced**; generation recipe in `figures/make_baseline_comparison.py`. (a) The friction the synthetic rollout can demand, (9): the baseline's 2–4 m/s clip reaches $\mu = 0.43$ on the full-scale vehicle but ≈ 2.0 on the 1:10 platform it was designed for, which is why the defect is scale-latent. (b) A known tire rolled out and fitted back across rollout speeds: inside the baseline clip the peak factor lands on its box bound, above 7 m/s it is recovered. (c) The $(\alpha_f, F_{y,f})$ data the Pacejka fit actually sees under each configuration, against the true curve; the baseline configuration supplies only the initial linear ramp, where solely the product BCD is identifiable.

exactly (Table VII); what is missing is the same comparison against the simulator's per-wheel forces across a pace sweep, including the σ_μ/μ gate boundary and the IMU-versus-kinematics cross-check that guards the mass and inertia assumptions.

5) Measure I_z rather than assuming it (Section A).

b) Beyond:

1) **Surface-change adaptation.** The bridge changes tire friction at runtime without respawning, making a step-change-in-grip experiment cheap: change μ mid-run and measure the identifier's convergence time and the controller's transient. This is the scenario on-track identification exists for, and it is difficult to stage on hardware.
2) **Excitation-aware data collection.** Equation (9) inverts into a requirement on the manoeuvre. A controller that deliberately inserts bounded slalom or ramp-steer excitation when the identifier reports insufficient slip-angle coverage would move the pipeline from passive on-track identification toward active, safety-bounded on-track experiment design.
3) **Uncertainty-aware output.** The gates of Section VII-A are binary. Reporting a posterior over the coefficients—the Bayesian variational path already implemented alongside the deterministic solvers—would let the controller weight the model by its confidence rather than accept or reject it.
4) **Hardware replication.** The decisive test of every claim in Section VIII is a physical Formula Student vehicle. The identification code is unchanged between simulation and hardware; what changes is the availability of ground truth, which is what makes the simulation study a necessary precursor rather than a substitute.

Appendix A
Assumptions and Reproducibility Notes

The following are assumptions or design choices rather than measured facts. They are listed explicitly so that a reader can judge how far each result depends on them, and so that each can be discharged by a specific measurement.

A1. Yaw inertia I_z

$I_z = 51.1\,\mathrm{kg\,m^2}$ is carried in the model file and is *not* measured. It enters (2) directly and therefore scales the yaw-rate residual the network is trained on, and it enters the stiffness-adaptive sub-stepping of Section VII-A through $l_f^2 C_f + l_r^2 C_r$ over I_z. *How to discharge:* apply a known yaw moment (a step steer at constant speed with the resulting axle forces read from the force feedback topic) and fit I_z from the measured $\dot{\omega}$; or read the value the physics engine is using directly from the actor's inertia tensor.

A2. Centre-of-gravity split

$l_f = 0.738142\,\mathrm{m}$, $l_r = 0.795362\,\mathrm{m}$ are taken from the controller configuration and confirmed by the vehicle owner. The measured static load split (51.2/48.8 front/rear) implies a marginally different split than the geometry does (51.9/48.1) — a 1.4 % disagreement. We have not resolved it, and it is below the level at which any conclusion in this paper would change.

A3. CG height and load transfer

$h_{\text{cg}} = 0.3\,\mathrm{m}$ is carried in the model file and unused: the single-track model of Section III-A neglects load transfer, following the baseline. Normal loads are static. On a full-scale vehicle at $1\,g$ this is a stronger approximation than at 1:10, and it is a candidate explanation for part of the residual the network absorbs.

A4. Steering-angle sign and units

`feedback/steering_angle` is published in degrees and is treated as the front road-wheel angle, positive left, matching the ROS body frame. It was verified to agree with the front-left wheel joint state to within 0.5 %. Left and right road wheels are not distinguished; the single-track δ is taken as this single reported value.

A5. Steady-state force recovery

Equation (7) is not a force measurement — it is fully determined by $v_x\omega$ — so it misattributes transients. It is applied only to the *synthetic*, quasi-steady rollout, where the steady-state assumption is constructed rather than hoped for. Its adequacy was nonetheless checked on measured data against the simulator's own axle force (correlation 0.993, slope 0.94); we treat that as validating the relation, not as validating a steady-state assumption on recorded laps.

A6. The residual network is not the plant

The corrected model $\hat{\mathbf{x}}_{k+1} + e_k$ is queried by the rollout at inputs the network has only partially seen. The extrapolation bound of Section V-C limits but does not eliminate this: the sweep still reaches $1.5\times$ the observed 99th percentile of $|\delta|$. The choice of 1.5 is empirical, selected because it was sufficient to reach the tire's peak at the corrected rollout speed while removing the observed drift; it is not derived.

A7. Simulator determinism and timing

CARLA runs in synchronous mode with a 0.033 s physics step, but sim time advances faster than wall time (observed $\approx 1.9\times$ on our hardware). All identification data is stamped from the simulation clock.

A8. Software configuration of record

Identification and control workspace: ROS 2 Humble, Python 3.10 (required — ROS 2 Humble ships CPython 3.10 ABI extensions only), PyTorch with CUDA on an RTX 2080-class GPU. The bridge is built and sourced as a separate overlay on top of that workspace, so that its message packages (`sim_manager_msgs`) resolve for the ground-truth forces; CARLA server 0.9.16 / Unreal Engine 4.26; **acados 0.5.5** with partial-condensing HPIPM. Identification runs at 20 Hz with a 30 s buffer, six co-identification iterations, 200 training epochs, full-batch Adam at 5×10^{-4}.

APPENDIX B
REFERENCE IMPLEMENTATION

The identification package, the ROS 2 bridge and the controller stack used in this paper are available as `On-Track-SysID` (extended from [6]), CARLA_ASU_BRIDGE [30], and the accompanying MPC and benchmarking packages. The benchmarking packages compute one-step and multi-step prediction metrics online (RMSE, MAE, NRMSE, MaxAE, bias, standard deviation and R^2 by Welford's algorithm), compare estimated against ground-truth per-wheel lateral forces, and record controller-switching behaviour, and were built for this study.

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The ON-TRACK-SYSID pipeline studied here originates from the ForzaETH Race Stack and the work of Dikici *et al.* [6], released under an open-source licence; this paper reports a port, a correction, and an extension of that work rather than a reimplementation from scratch.

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